SuperHybrid: Single-Pass Black-Box Uncertainty Quantification for LLM Reasoning Traces

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Abstract

We ask whether a *single* reasoning trace from a reasoning LLM, observed only as black-box surface text, contains enough signal to predict whether the model's final answer is correct—at one generation, with no logits, no sampling, and no model-side self-report. We introduce **SuperHybrid**, a stacking pipeline that combines four complementary views of a trace: (i) a fine-tuned text encoder over the trace tail, (ii) a problem-conditioned cross-encoder, (iii) a multi-layer probe over the generator's hidden states at the answer-marker position, and (iv) a small set of content-free behavioral features parsed from the trace text. Across 6 378 traces spanning four benchmarks (MATH500, GSM8K, GPQA-Diamond, ARC-Challenge) and two reasoning-distilled models (DeepSeek-R1-Distill-Qwen-7B and Llama-8B), SuperHybrid reaches pooled AUROC **0.815** (LR meta-learner) **/ 0.807** (RF) with expected calibration error **0.042**, beating a fine-tuned DeBERTa-v3 read of the trace by $+0.05$ **AUROC** (paired DeLong $p < 10^{-19}$) and halving its ECE. On MATH500-Qwen7B alone the system reaches **0.934** AUROC. An ablation across 328 additional handcrafted structural features (n-gram motifs, graph topology, timing, persistent homology) adds ≤ 0.005 AUROC—a *text-saturation* result that informs the recipe: a small text + probe + random-forest stack is enough.

1 Introduction

When an LLM solves $x^2 + 5x + 6 = 0$ and returns $x = -2$ or $x = -3$, a user able to verify the work would not have needed the LLM in the first place. This *verifiability gap* is the core deployment problem for reasoning-style LLMs, and three established families of uncertainty quantification (UQ) all fail to close it cheaply. **Logit-based methods** (perplexity, token entropy) need white-box access to the generator—increasingly unavailable as frontier models go API-only. **Sampling methods** (self-consistency, semantic entropy) require $N \geq 5$ independent generations per query, multiplying inference cost 5–20×—prohibitive when reasoning traces already span thousands of tokens. **Verbalized confidence** ("How confident are you, 0–100?") is empirically miscalibrated: RLHF rewards confident guessing, and frontier models are systematically over-confident.

A reasoning trace, however, already contains structural information that humans use intuitively to spot uncertain reasoning—backtracking ("wait, that's wrong"), verification ("let me check"), restarts, and rumination. Marjanović et al. [**?**] document on AIME-24 that *correct* DeepSeek-R1 traces average $\approx 2{,}000$ tokens whereas *incorrect* ones average $\approx 4{,}000$, and rumination rate is negatively correlated with accuracy. Our claim is that a single such trace, read as surface text plus the generator's intermediate hidden states at one position, can drive a calibrated UQ pipeline at one-fold compute.

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Contributions. (i) We propose **SuperHybrid**, a four-signal stacked predictor of single-trace correctness that operates at one generation, requires only the trace text and a single hidden-state slice, and matches or beats sampling-based UQ at a fraction of the cost. (ii) We report pooled AUROC 0.815 with ECE 0.042 across 4 benchmarks $\times\,2$ models, with a peak of 0.934 on MATH500-Qwen7B and a $+0.05$ AUROC lift over a fine-tuned DeBERTa-v3 baseline. (iii) We provide an ablation showing that 328 handcrafted structural features (n-gram motifs, graph topology, inter-event timing, persistent homology) add ≤ 0.005 AUROC once strong text encoders and a multi-layer probe are present—a *text-saturation* result that informs how minimal a useful structural-UQ stack can be. (iv) We demonstrate cross-domain transfer (MATH500 $\rightarrow$ GPQA-Diamond) with AUROC 0.73, showing that the signal generalizes beyond the training domain when trace shapes are similar.

2 Related Work

Topology of reasoning. Minegishi et al. [**?**] cluster the hidden activations of a reasoning model and study the resulting reasoning *graph*—cyclicity, diameter, small-world index. Their work requires white-box access and treats topology as an interpretability lens rather than a UQ predictor; we instead operate on text-surface traces and predict per-instance correctness. Persistent-homology work on traces and Da et al. [**?**] target *explanation* or quality assessment rather than answer correctness. Gandhi et al. [**?**] define the six-way cognitive behavior taxonomy (forward, verify, revise, restart, hesitate, conclude) we adopt; [**?**] provides the empirical link between behavior frequencies and accuracy.

Single-pass black-box UQ. Verbalized confidence [**?**] and trace-length heuristics are known to miscalibrate. Sampling baselines—self-consistency [**?**] and semantic entropy [**?**]—require $N \geq$ 5–8 generations per query; LM-Polygraph [**?**] consolidates these for benchmarking. Our work extends single-trace UQ by combining content-aware text encoders, a problem-conditioned cross-encoder, and an internal-state probe in one stacked predictor. Probing intermediate transformer activations as a UQ predictor is motivated by the empirical observation that mid-late layers hold the most generalization-relevant features.

3 Method

3.1 Four complementary signals

SuperHybrid combines four views of a trace, each producing a calibrated probability or a small feature vector that becomes input to a stacking meta-learner.

(i) Text surface, $P_{\textbf{text}}$. A fine-tuned DeBERTa-v3-base classifier on the last 512 tokens of the trace (“trace tail”), 5-fold stratified cross-validation. The trace tail concentrates the model’s final consolidation step.

(ii) Problem alignment, $P_{\textbf{cond}}$. A problem-conditioned cross-encoder DeBERTa reading (problem $\|$ trace tail). Captures topic drift—when the trace becomes inconsistent with the original question.

(iii) Internal state, $P_{\textbf{probe}}$. A small MLP probe over the generator’s hidden states at the answer-marker position, concatenated across four decoder layers ($\approx 25, 50, 75, 100\%$ depth). Empirically the mid-late layers ($\approx 71\%$ depth) carry the cleanest signal, consistent with the probing literature.

(iv) Behavioral features, F. A 28-dim vector parsed using the six-class behavior taxonomy (forward / verify / revise / restart / hesitate / conclude): length and proportions, structural features (verification-to-forward ratio, backtrack position, transition entropy, behavior cycle count, longest forward run), and content-free meta-features (wait-density, question-mark density, repetition rate).

3.2 Stacking with leakage-safe out-of-fold predictions

Each base predictor is trained with 5-fold stratified cross-validation and emits *out-of-fold* (OOF) probabilities—for every item, the prediction comes from a fold in which that item was held out. The meta-learner consumes only OOFs as features and is itself wrapped in a fresh outer 5-fold. Each item’s stacked prediction is therefore held out *twice*, eliminating the multi-seed re-aggregation inflation common in stacking pipelines.

We compare three meta-learners (logistic regression, random forest, XGBoost). Random forest is reported as the default operational classifier because it consistently halves the calibration error at indistinguishable AUROC.

Evaluation

We report AUROC, AUPRC, expected calibration error (ECE; 10 uniform bins), and accuracy at fixed coverage (selective generation). Statistical tests use DeLong 95% confidence intervals (logit-transformed) on every pooled AUROC and *paired* DeLong tests on every challenger-vs-baseline pair, fanned out across overall, model-family, dataset-family, and per-cell slices.

4 Experimental Setup

Datasets and generation. We use four publicly released test splits: MATH500 [**?**] (a 500-item subset of the MATH competition-mathematics benchmark), GSM8K [**?**] (1,319 grade-school word problems), GPQA-Diamond [**?**] (198 graduate-level science MCQ), and ARC-Challenge [**?**] (1,172 science MCQ). Each item is solved by both DeepSeek-R1-Distill-Qwen-7B and Llama-8B [**?**] (open-weight, MIT and Llama Community licenses) at $T = 0.6$, top-$p = 1$, max 32,768 tokens, exactly one sample per prompt — 6,378 traces across 8 dataset×model cells. Final answers are extracted from outside the <think> block and scored by exact-match (math) or letter-match (MCQ); no other preprocessing is applied to the trace text.

Behavior parser. Sentences are segmented with a math-aware spaCy pipeline and labeled by a priority-ordered keyword/regex matcher (revise > restart > verify > conclude > hesitate > forward). Inter-annotator agreement against 100 manually-labeled traces is Cohen's $\kappa = 0.74$.

Compute and baselines. Trace generation, hidden-state extraction, and DeBERTa fine-tuning ran on Yale University's Grace HPC cluster (NVIDIA A100/H100 GPUs allocated via SLURM, PyTorch 2.2); behavioral-feature parsing, stacking, and statistical analysis are CPU-only on a workstation. Hyperparameters and the Optuna tuning protocol are detailed in Appendix **??**. We compare SuperHybrid against (a) length-only logistic regression; (b) seven lexical surface cues; (c) the 28 handcrafted behavioral features alone; (d) TF-IDF unigram+bigram logistic regression; (e) plain DeBERTa-v3-base fine-tuned on the trace tail—the text-encoder bar a structural-UQ method must clear to claim that structure adds value over content.

5 Results

5.1 Pooled AUROC and calibration

Table **??** reports pooled AUROC and ECE across the eight cells (n = 6,344–6,378 depending on OOF coverage). 95% DeLong confidence intervals are shown for the most relevant comparators.

Table 1: Pooled AUROC and ECE across 8 dataset×model cells.

Method	AUROC	95% CI	ECE
Length-only LR	0.595	—	—
Lexical cues LR (7 feats)	0.628	—	—
Behavioral features alone (28)	0.666	[0.652, 0.679]	—
TF-IDF LR (≈20k feats)	0.749	—	—
Plain DeBERTa-v3, trace tail	0.762	[0.751, 0.774]	0.090
Problem-conditioned DeBERTa	0.788	[0.777, 0.799]	0.086
Multi-layer probe (alone)	0.776	[0.764, 0.787]	—
Mean of three probabilities	0.805	[0.792, 0.813]	0.082
SuperHybrid (LR meta-learner)	**0.815**	**[0.804, 0.826]**	0.080
SuperHybrid (RF meta-learner)	**0.807**	**[0.797, 0.818]**	**0.042**

SuperHybrid beats the plain DeBERTa text-encoder bar by +0.045 (RF) to +0.053 (LR) AUROC, paired DeLong $p < 10^{-19}$. The RF variant *also* halves ECE from 0.090 to 0.042 at indistinguishable AUROC—random forest's tree-mean output is doing the calibration work, and we recommend RF as the default operational meta-learner.

5.2 Per-cell results and cross-domain transfer

Table **??** (left) shows AUROC of SuperHybrid (RF) per cell, strongest on math; Qwen cells are uniformly ≈ 0.06 above Llama, an asymmetry that persists across every base predictor. The right panel reports out-of-domain AUROC when we train RF on MATH500 features and evaluate without retraining: MATH500 $\rightarrow$ GPQA-Diamond transfers at 0.73, *above* the length-only baseline (0.65)—the signal is genuinely structural, not length-driven. Transfer fails to ARC and GSM8K (≈ 0.51–0.58), whose trace shapes (short MCQ rationales, grade-school arithmetic chains) differ markedly from MATH500's competition-mathematics traces.

Table 2: (Left) per-cell AUROC, SuperHybrid (RF). (Right) out-of-domain AUROC.

Dataset	Qwen-7B	Llama-8B
MATH500	**0.934**	0.868
GSM8K	0.860	0.766
GPQA-Diamond	0.768	0.697
ARC-Challenge	0.733	0.683

Target	n	RF	Length-only
GPQA-Diamond	198	**0.732**	0.645
ARC-Challenge	1172	0.577	0.584
GSM8K	1319	0.508	0.492

5.3 What does and does not contribute: text-saturation

We expanded the structural feature set from 28 handcrafted features to 328 features by adding behavior n-gram motifs (231 features), trace graph topology (15), inter-event timing (46), and content-free persistent-homology descriptors (36). Stacking the 328-feature block on top of the four signals in Section 3.1 yields the leave-one-family-out deltas in Table **??**.

Table 3: Ablation. ΔAUROC of removing each family from the all-in stack (positive = contributes, negative = hurts).

Removed family	ΔLR	ΔRF	ΔXGB
Text encoders (DeBERTa, conditioned, RoBERTa)	$+0.036$	$+0.043$	$+0.036$
Multi-layer probe	$+0.010$	$+0.012$	$+0.011$
328 expanded structural features	-0.021	-0.005	$+0.002$

Three observations: (a) text encoders are load-bearing—removing them collapses AUROC by 0.04, four times any other family; (b) the multi-layer probe contributes a uniform $+0.01$, the only consistent non-text gain; (c) expanding the behavioral block from 28 to 328 features adds zero or negative value once text encoders and the probe are present (LR is *hurt* by -0.021). We interpret (c) as **text-saturation**: the lexical signatures of revision ("wait, that's wrong"), verification ("let me check"), and restart ("starting over") that drive structural-feature counts also feed the text encoder directly. Handcrafted features remain useful for *interpretability* and pass a length-control test (they beat length-only inside every length quintile, ΔAUROC $= 0.04$–0.08 per bin), but add no stacking-level signal beyond the small probe gain.

6 Discussion

Calibration and the minimal recipe. LR sits near ECE 0.080 and RF near 0.042; the AUROC distinction is within paired-DeLong noise ($\Delta \approx 0.005$), but the calibration gap is two-fold and structural—RF's tree-mean averaging halves it. Combined with text-saturation, this argues for a deliberately *minimal* stack: trace-tail encoder + problem-conditioned encoder + multi-layer probe + small behavioral vector, stacked with RF. The recipe is computationally light—one generation plus one teacher-forcing pass. At 70% coverage on MATH500, SuperHybrid answers $\approx 95\%$ of kept items correctly, competitive with self-consistency $N = 8$ ($\approx 92\%$) at one-eighth the cost.

Limitations, future work, and takeaway. The system applies only to open-source models and verifiable benchmarks; a persistent Qwen $\leftrightarrow$ Llama asymmetry of ≈ 0.06 AUROC remains unexplained. Future directions include running the probe at every k-th token for mid-trace abort, distilling it into a text-only student to remove the white-box requirement, and extending to medical, legal, and code reasoning. Overall, SuperHybrid shows that calibrated single-pass UQ from one reasoning trace is achievable, and that the right recipe is a deliberately minimal text + probe + RF stack.

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A Computing Infrastructure and Hyperparameters

Compute resources. All trace generation, hidden-state extraction, and DeBERTa fine-tuning were performed on Yale University's *Grace* HPC cluster, with jobs scheduled via SLURM on NVIDIA B200 and H200 GPUs from the gpu$_{200}and$

We used Anthropic's Claude (Sonnet 4.6 / Opus 4.7, accessed via Claude Code) as a coding and writing assistant during this project. We disclose its use here in the spirit of the course's AI-use policy.

Where AI helped.

- Code review and debugging. Diagnosing a numpy ragged-array bug (ValueError: could not broadcast (2,N) into (2,)) in the trace-DAG construction script; identifying a misalignment in the behavior taxonomy mapping between the legacy and new parsers (the legacy 7-class enum lacked a REVISE member, silently mapping revise events to PAD). Both fixes were applied by the authors after AI-assisted diagnosis.
- Statistical analysis. Interpreting paired DeLong test results across the dozens of challenger-vs-baseline comparisons we ran -- which ΔAUROC's are statistically significant given sample size and relative variance; which apparent per-slice wins/losses are noise.
- Report drafting and revision. Generating initial drafts of this report, restructuring sections, compressing for the 4-page NeurIPS limit, formatting tables, and producing the LaTeX source. The first author iterated heavily over each draft and made all final content decisions.
- Architectural sounding-board. Discussing whether to drop or retain individual signal families, what ablations would be informative, and how to frame the negative results we encountered (e.g., the 328-feature behavioral block adding zero AUROC at the meta-stack level).

Where AI did NOT contribute.

- Core algorithmic code (the multi-layer probe, behavioral parser, graph encoders, feature extractors, OOF stacking infrastructure) was written by the authors.
- No experimental number in this paper was generated by an LLM. All AUROCs, ECEs, p-values, confidence intervals, and per-cell breakdowns come from actual model runs and DeLong computations on real data, recorded in the project's results/ and reports/ directories.
- Final scientific judgments -- which results to highlight, what claims to make, what limitations to acknowledge, what counts as a falsifier -- are the authors'.

B Representative Prompts

A small selection of the prompts we issued to Claude during the report-drafting and analysis phase (lightly normalized for layout; full session logs are retained on the project server).

1. *Conclusion request.* "Let's now conclude this research with a final report ($\sim$ 3 pages). The report should demonstrate the following aspects in detail: (1) Motivation and problem statement, (2) Related work and literature review, (3) Results and evaluation, (4) Discussion and analysis. Negative results will not be penalized but should be accompanied with detailed and comprehensive analysis of why the proposed methods didn't work and provide some additional insights into the problem."
2. *Revision request.* "This is meant to be a short conference paper, and the readers don't know what our internal experiment labels are. They should not appear in the paper; where you reference them, explain what they are clearly. We don't need to mention experiments that were not carried out, and we can skip the pre-committed

falsifiers (those were just pre-research thoughts). Focus on what we achieved actually."

3. *LaTeX porting.* "Create the LaTeX version of our report following this NeurIPS format, and make sure it's within 4 pages (references and appendix do not count toward the page limit)."

4. *Statistical interpretation.* "I just got the new paired DeLong results and synced them. Please meticulously examine, think, and reason through each of them step by step in detail. Then demonstrate what they mean exactly and what the implications are."

5. *Diagnosis.* "I received this error from the trace-DAG build: [traceback]. Please diagnose and address the error step by step in detail."

6. *Compliance check.* "Here is a checklist for the report the professor gave us. Please meticulously think and reflect through our final report again to make sure it satisfies at least all the checklist items. If there's anything missing, revise and enhance our final report accordingly."